

AES-128 CRYPTO ACCELERATOR SoC

RTL-to-GDSII on Sky130 | 227x faster than software | NIST FIPS-197 verified

Akif Mohamed (Physical Design) - Nandikha (RTL / Synthesis / STA) | github.com/akifmohamed/AES-Crypto-SoC

OVERVIEW

- AES-128 encryption accelerator SoC (iterative 1-round core + UART SoC), 11 cycles @ 50 MHz = 220 ns per block vs ~50,000 ns software on MCU = 227.3x speedup.
- Hand-written Verilog (15 files), full RTL-to-GDSII with Yosys / OpenSTA / OpenLane 2 on Sky130; layout reviewed in licensed Cadence Virtuoso 6.1.5.

SILICON & TIMING RESULTS

- Die 1000x1000 um (1 mm²); utilization 45% planned, 20.2% achieved (162,993 um²) - die sized for first-pass closure; ~1.8 m total wire, 373K vias.
- Yosys synthesis (sky130_fd_sc_hd): 185,254 um², 8.3% sequential (S-Box LUT dominance); ShiftRows = 0 cells (pure wiring).
- Timing closure @ 20 ns / 50 MHz: worst setup +14.07 ns, worst hold +0.29 ns (nom_tt corner).
- Signoff: DRC clean, LVS MATCH, antenna repair enabled; 64 MB GDSII versioned via Git LFS.

PHYSICAL DESIGN OWNERSHIP (AKIF)

- Floorplan (die/core, power rings & straps) -> placement -> CTS -> routing -> signoff in OpenLane 2.3.10.
- Rehearsal-run methodology de-risked the official spin; flow resumed from last saved step after a mid-run power cut; OpenLane-1 config keys migrated (RUN_ANTENNA_REPAIR).

DEBUGGING HIGHLIGHTS

- Found -38 ns structural critical path post-CTS; proved it buffering-insensitive over 3,300 resizer iterations; drove RTL fix (iterative 1-round pipeline) that closed timing next spin.
- Fixed OpenSTA-incompatible SDC (Synopsys-only set_fix_hold removed; hold repaired inside PNR).
- Solved PDK-less Virtuoso XStream import (read-only-tech error) with a custom Sky130 GDS layer map.

VERIFICATION, DEMO & SIGNOFF VIEWING

- 5/5 NIST FIPS-197 vectors (TV1 cipher 3AD77BB40D7A3660A89ECA32466EF97; FPGA LED byte 0x97).
- pc_demo: SW 50,000 ns vs HW 220 ns = 227.3x; Basys3/iCEstick constraints shipped, bitstream board-ready.
- Virtuoso screenshot set (licensed signoff viewing) + KLayout color views (open-PDK layer map) - standard mixed-flow practice.

GROWTH STORY

- Follow-up to PipeCore-GDS (8-bit pipelined ALU, 168 cells, DRC/LVS clean): 50x design complexity, 122x silicon area, full PD ownership across both RTL-to-GDSII projects.